

Instrumentation

Temperature Sensors Cont.

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فروردین ماه 1404

- جبران سازی دمایی در ترموکوپل ها
- حسگرهای دمایی مدار مجتمع
 - اصول کارکرد
 - مثالهایی از IC های مدار مجتمع

روش ۱: استفاده از یک حسگر اضافی

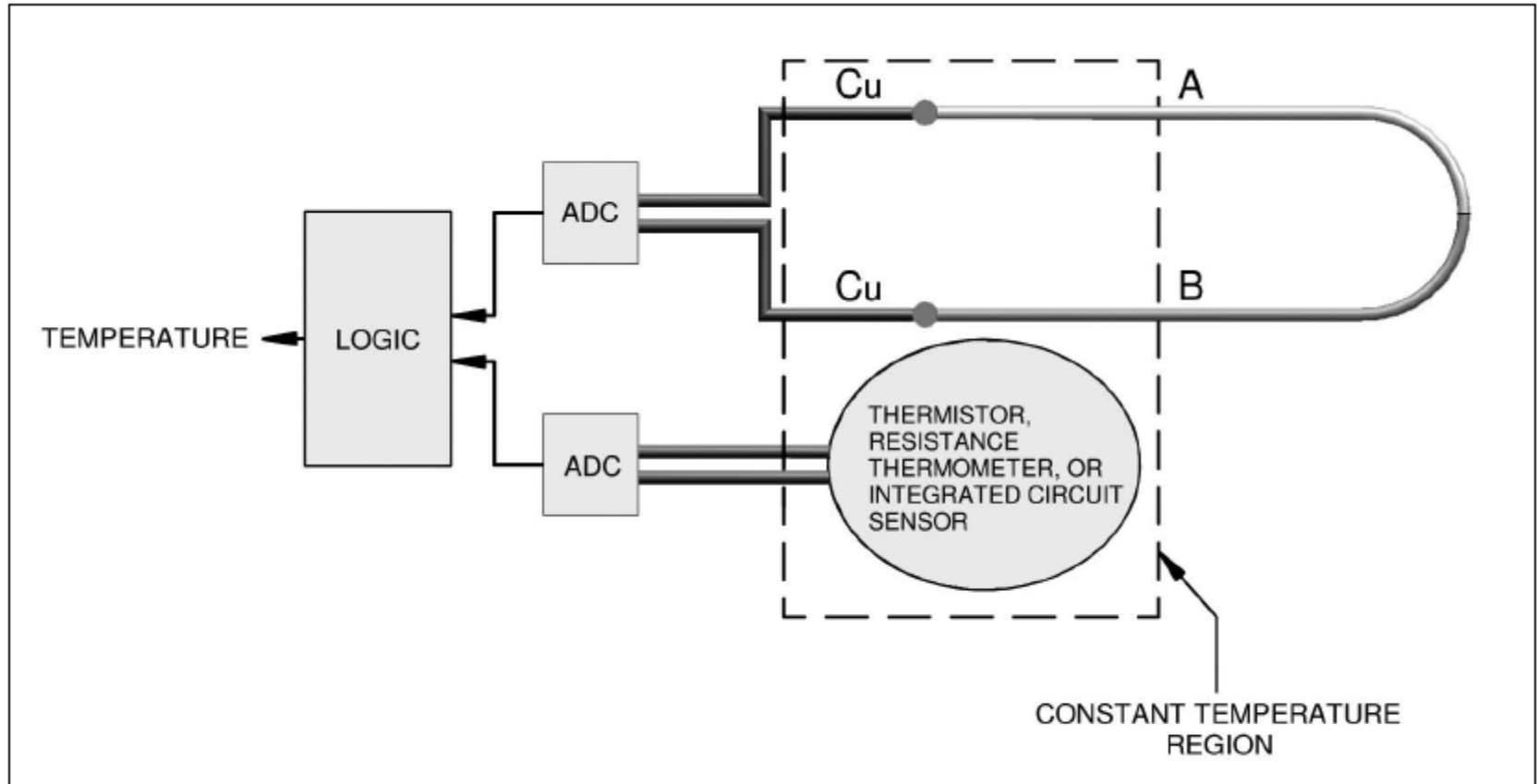


Figure 3-3. Computational Reference Junction Compensation

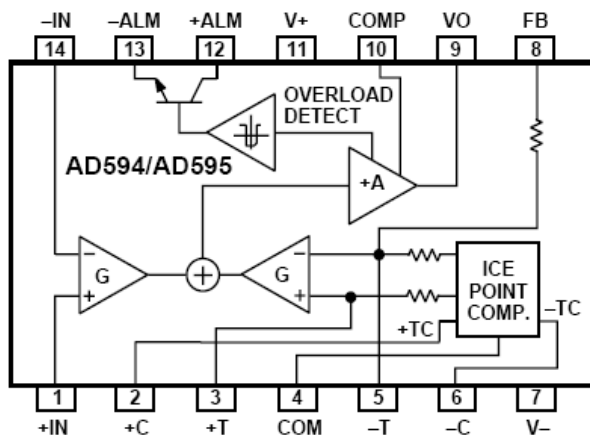
روش ۲: استفاده از مدارهای مجتمع خاص

Monolithic Thermocouple Amplifiers with Cold Junction Compensation

AD594/AD595

FEATURES

- Pretrimmed for Type J (AD594) or Type K (AD595) Thermocouples
- Can Be Used with Type T Thermocouple Inputs
- Low Impedance Voltage Output: 10 mV/°C
- Built-In Ice Point Compensation
- Wide Power Supply Range: +5 V to ± 15 V
- Low Power: <1 mW typical
- Thermocouple Failure Alarm
- Laser Wafer Trimmed to 1°C Calibration Accuracy
- Setpoint Mode Operation
- Self-Contained Celsius Thermometer Operation
- High Impedance Differential Input
- Side-Brazed DIP or Low Cost Cerdip



The temperature sensitivity of silicon integrated circuit transistors is quite predictable and repeatable. This sensitivity is exploited in the AD594/AD595 to produce a temperature related voltage to compensate the reference of “cold” junction of a thermocouple as shown in Figure 16.

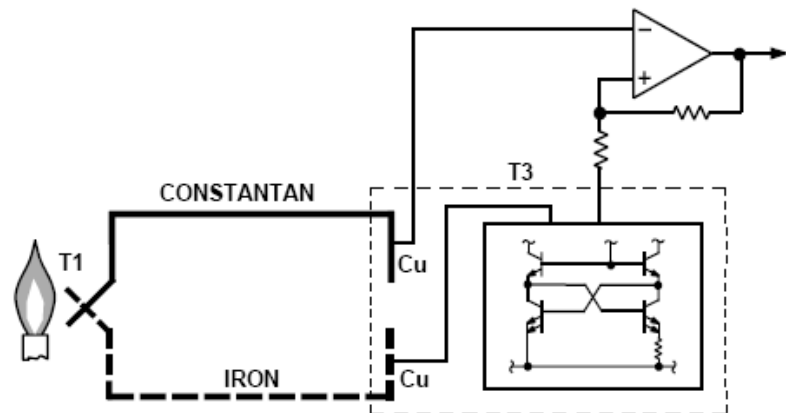


Figure 16. Connecting Isothermal Junctions

استفاده از مدارهای مجتمع AD594/AD595 برای جبران سازی دمایی و تقویت

Table I. Output Voltage vs. Thermocouple Temperature (Ambient +25°C, $V_s = -5\text{ V}, +15\text{ V}$)

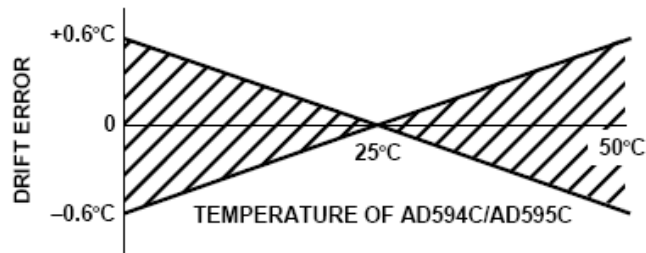


Figure 8. Drift Error vs. Temperature

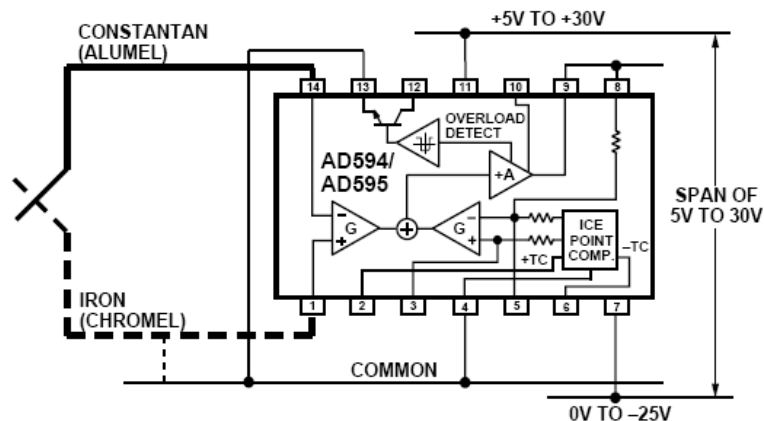


Figure 2. Dual Supply Operation

Thermocouple Temperature °C	Type J Voltage mV	AD594 Output mV	Type K Voltage mV	AD595 Output mV
-200	-7.890	-1523	-5.891	-1454
-180	-7.402	-1428	-5.550	-1370
-160	-6.821	-1316	-5.141	-1269
-140	-6.159	-1188	-4.669	-1152
-120	-5.426	-1046	-4.138	-1021
-100	-4.632	-893	-3.553	-876
-80	-3.785	-729	-2.920	-719
-60	-2.892	-556	-2.243	-552
-40	-1.960	-376	-1.527	-375
-20	-.995	-189	-.777	-189
-10	-.501	-94	-.392	-94
0	0	3.1	0	2.7
10	.507	101	.397	101
20	1.019	200	.798	200
25	1.277	250	1.000	250
30	1.536	300	1.203	300
40	2.058	401	1.611	401
50	2.585	503	2.022	503
60	3.115	606	2.436	605
80	4.186	813	3.266	810
100	5.268	1022	4.095	1015
120	6.350	1233	4.910	1210

استفاده از مدارهای مجتمع AD594/AD595

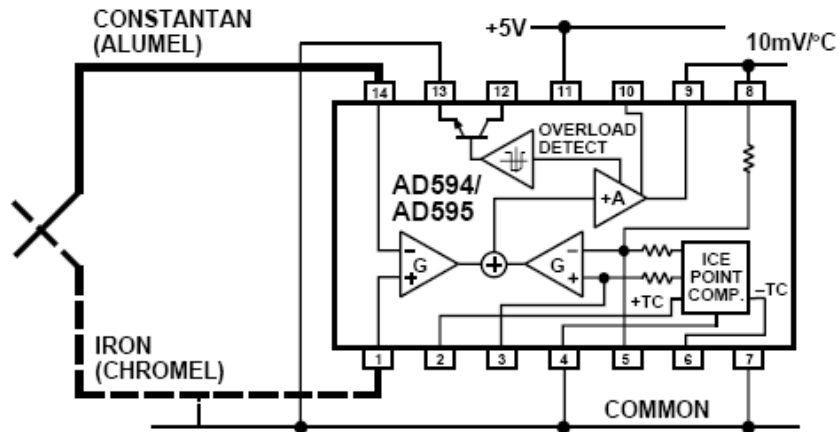


Figure 1. Basic Connection, Single Supply Operation

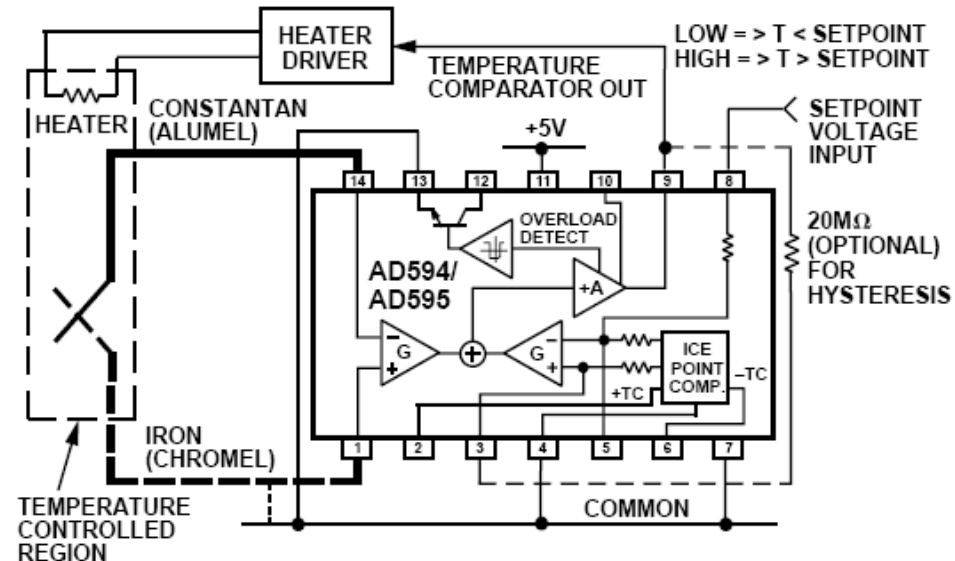


Figure 9. Setpoint Controller

Hysteresis can be introduced by injecting a current into the positive input of the feedback amplifier when the output is toggled high. With an AD594 about 200 nA into the +T terminal provides 1°C of hysteresis.

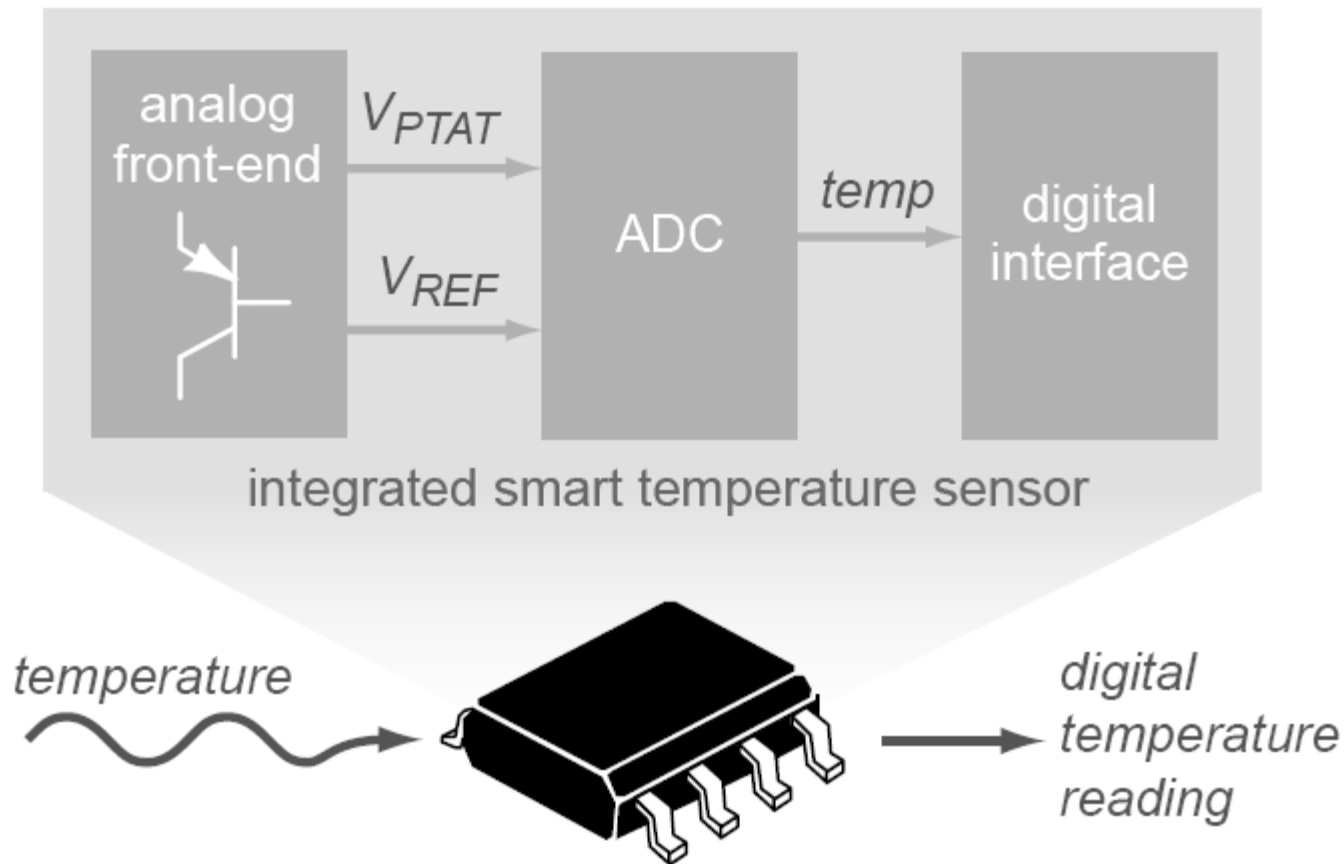
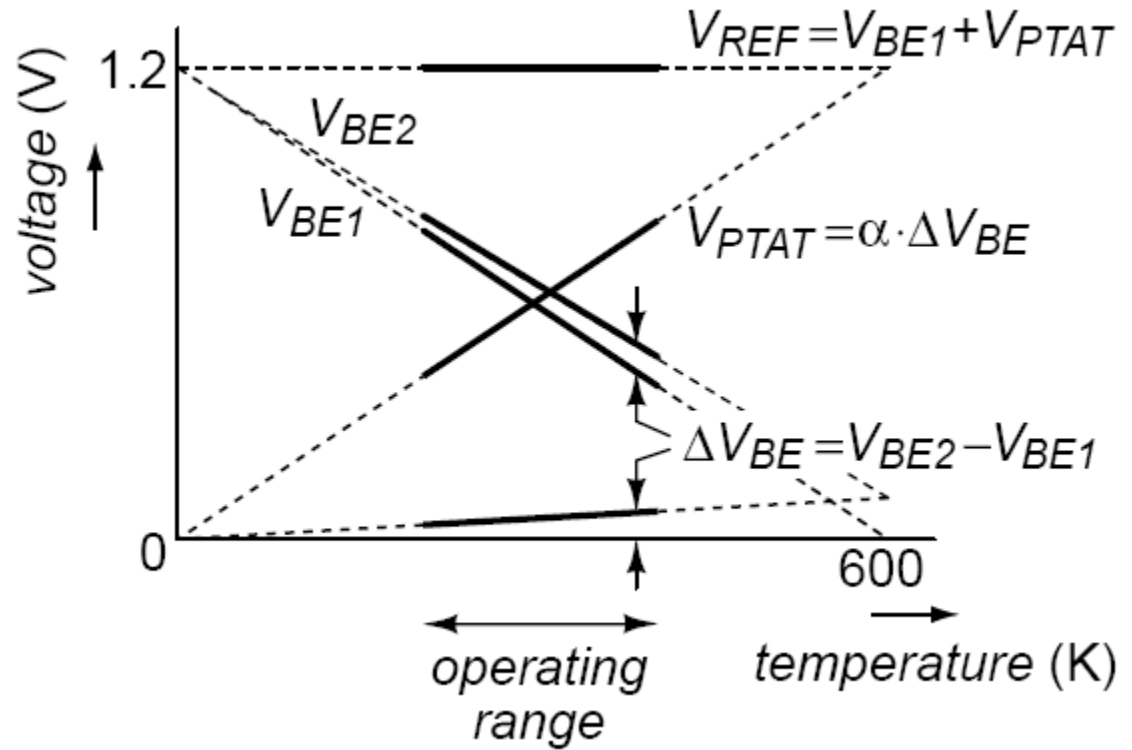
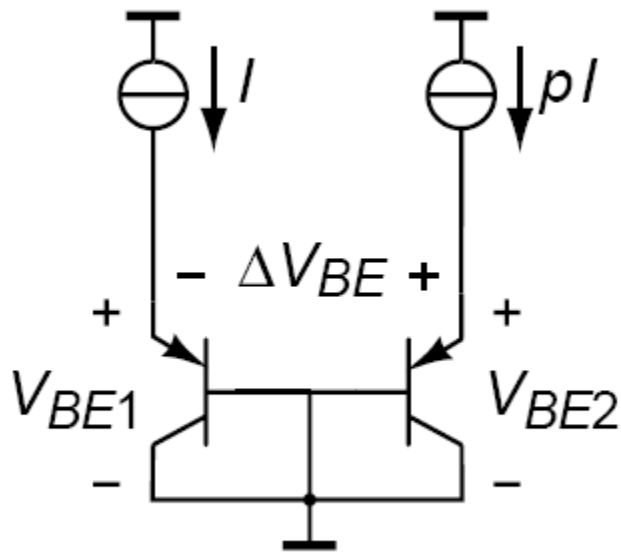


Figure 1.1. Block diagram of an integrated smart temperature sensor.



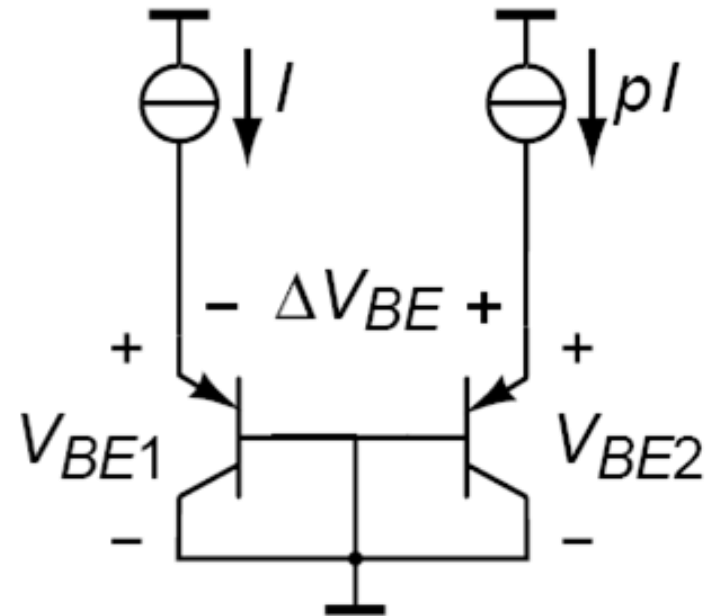
Proportional to Absolute Temperature (PTAT)

If an external voltage V is applied to a p-n junction, the balance between the drift and diffusion currents is disturbed:

$$I_D = I_S \left(\exp \left(\frac{qV}{kT} \right) - 1 \right)$$

For forward-bias voltages $V \gg kT/q$,

$$V = \frac{kT}{q} \ln \left(\frac{I}{I_S} \right)$$

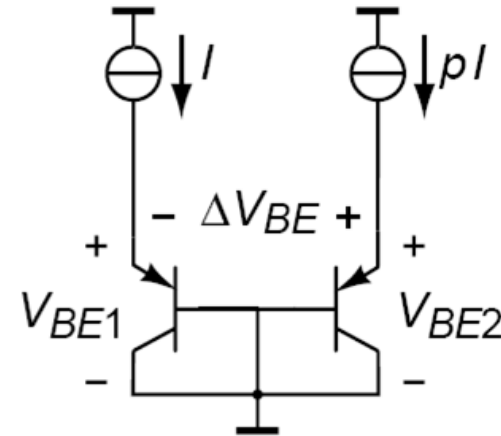


For forward-bias voltages $V \gg kT/q$,

$$V = \frac{kT}{q} \ln \left(\frac{I}{I_S} \right)$$

If two bias currents I_1 and $I_2 = pI_1$ are successively applied to a diode,

$$V_2 - V_1 = \frac{kT}{q} \ln \left(\frac{pI_1}{I_S} \right) - \frac{kT}{q} \ln \left(\frac{I_1}{I_S} \right) = \frac{kT}{q} \ln p.$$



اختلاف ولتاژهای بیس-امیترها نسبت به دما، خطی است.

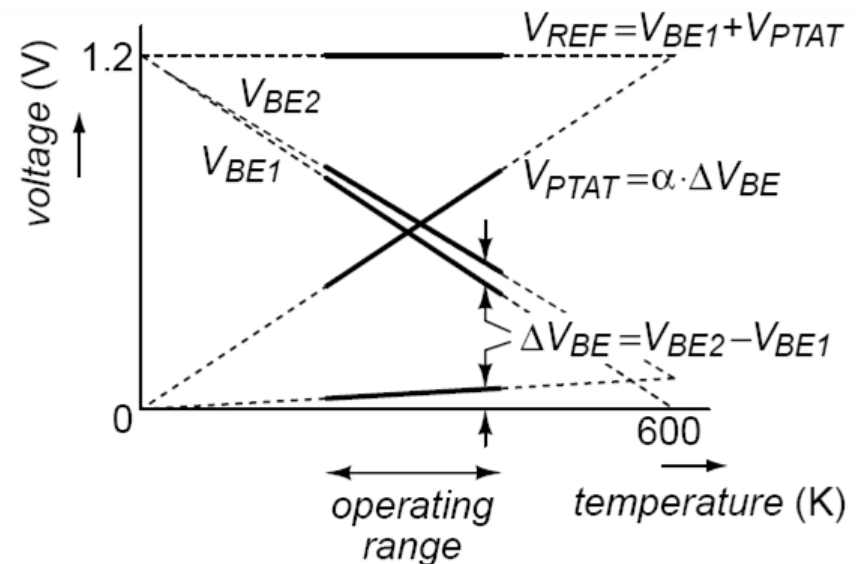


Table 7.6. Accuracy comparison of smart temperature sensors.

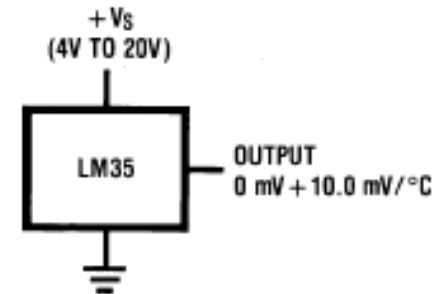
Reference	Inaccuracy	Range
LM75 [1]	$\pm 2.0^{\circ}\text{C}$	-25°C to 100°C
	$\pm 3.0^{\circ}\text{C}$	-55°C to 125°C
LM92 [17]	$\pm 0.33^{\circ}\text{C}$	at 30°C
	$\pm 1.5^{\circ}\text{C}$	-25°C to 150°C
DS1626 [18],	$\pm 0.5^{\circ}\text{C}$	0°C to 70°C
	$\pm 2.0^{\circ}\text{C}$	-55°C to 125°C
ADT7301 [19]	$\pm 1.0^{\circ}\text{C}$	0°C to 70°C
	$\pm 3.0^{\circ}\text{C}$	-40°C to 125°C
SMT160-30 [20]	$\pm 0.7^{\circ}\text{C}$	-30°C to 100°C
	$\pm 1.2^{\circ}\text{C}$	-45°C to 130°C

LM35

Precision Centigrade Temperature Sensors

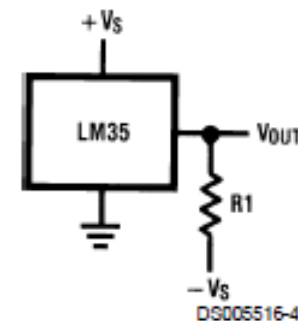
Features

- Calibrated directly in ° Celsius (Centigrade)
- Linear + 10.0 mV/°C scale factor
- 0.5°C accuracy guaranteeable (at +25°C)
- Rated for full -55° to +150°C range
- Suitable for remote applications
- Low cost due to wafer-level trimming
- Operates from 4 to 30 volts
- Less than 60 μ A current drain
- Low self-heating, 0.08°C in still air
- Nonlinearity only $\pm 1/4^\circ\text{C}$ typical
- Low impedance output, 0.1 Ω for 1 mA load



DS005516-3

FIGURE 1. Basic Centigrade Temperature Sensor (+2°C to +150°C)



DS005516-4

Choose $R_1 = -V_S/50 \mu\text{A}$
 $V_{OUT} = +1,500 \text{ mV at } +150^\circ\text{C}$
 $= +250 \text{ mV at } +25^\circ\text{C}$
 $= -550 \text{ mV at } -55^\circ\text{C}$

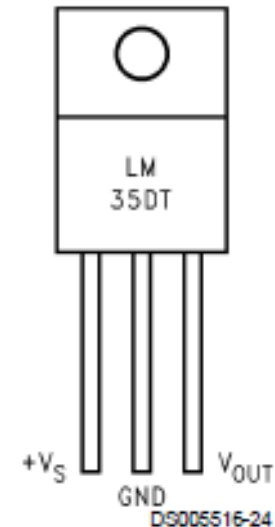
FIGURE 2. Full-Range Centigrade Temperature Sensor

Absolute Maximum Ratings (Note 10)

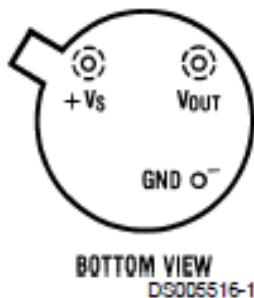
If Military/Aerospace specified devices are required, please contact the National Semiconductor Sales Office/ Distributors for availability and specifications.

Supply Voltage	+35V to -0.2V
Output Voltage	+6V to -1.0V
Output Current	10 mA
Storage Temp.:	
TO-46 Package,	-60°C to +180°C
TO-92 Package,	-60°C to +150°C
SO-8 Package,	-65°C to +150°C
TO-220 Package,	-65°C to +150°C
Lead Temp.:	
TO-46 Package,	
(Soldering, 10 seconds)	300°C

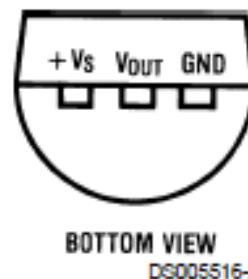
TO-220
Plastic Package*



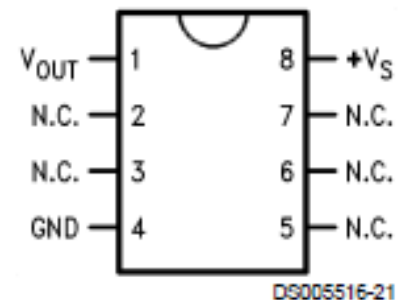
TO-46
Metal Can Package*



TO-92
Plastic Package



SO-8
Small Outline Molded Package

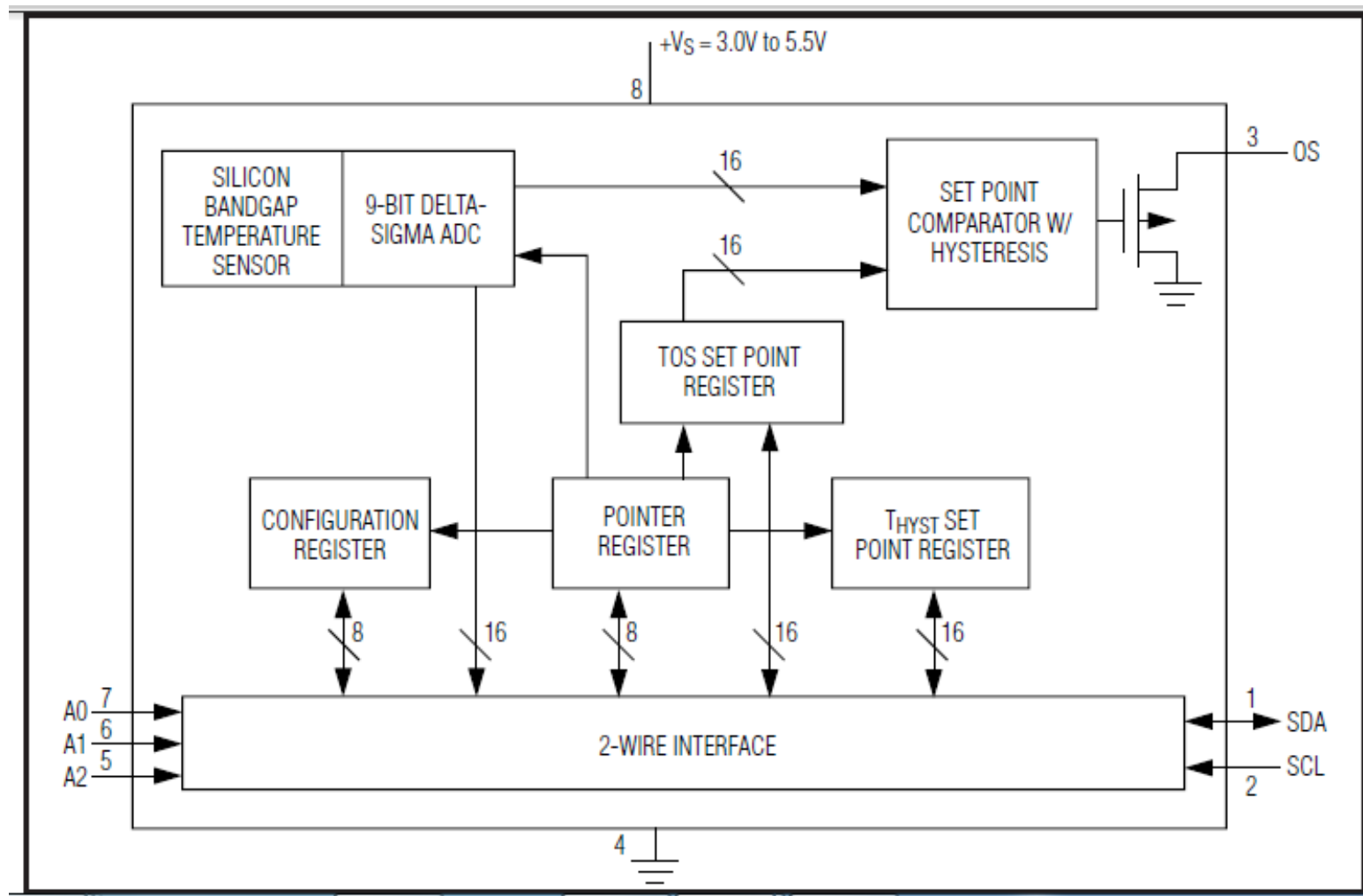


Electrical Characteristics

(Notes 1, 6)

Parameter	Conditions	LM35A		
		Typical	Tested Limit (Note 4)	Design Limit (Note 5)
Accuracy (Note 7)	$T_A = +25^{\circ}\text{C}$ $T_A = -10^{\circ}\text{C}$ $T_A = T_{\text{MAX}}$ $T_A = T_{\text{MIN}}$	± 0.2 ± 0.3 ± 0.4 ± 0.4	± 0.5 ± 1.0 ± 1.0	
Nonlinearity (Note 8)	$T_{\text{MIN}} \leq T_A \leq T_{\text{MAX}}$	± 0.18		± 0.35
Sensor Gain (Average Slope)	$T_{\text{MIN}} \leq T_A \leq T_{\text{MAX}}$	+10.0	+9.9, +10.1	
Load Regulation (Note 3) $0 \leq I_L \leq 1 \text{ mA}$	$T_A = +25^{\circ}\text{C}$ $T_{\text{MIN}} \leq T_A \leq T_{\text{MAX}}$	± 0.4 ± 0.5	± 1.0	± 3.0
Line Regulation (Note 3)	$T_A = +25^{\circ}\text{C}$ $4\text{V} \leq V_S \leq 30\text{V}$	± 0.01 ± 0.02	± 0.05	± 0.1

Digital Temperature Sensor and Thermal Watchdog with 2-Wire Interface



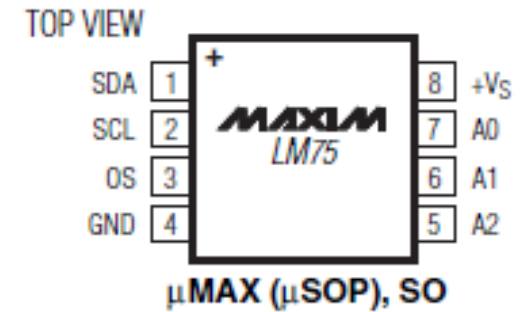
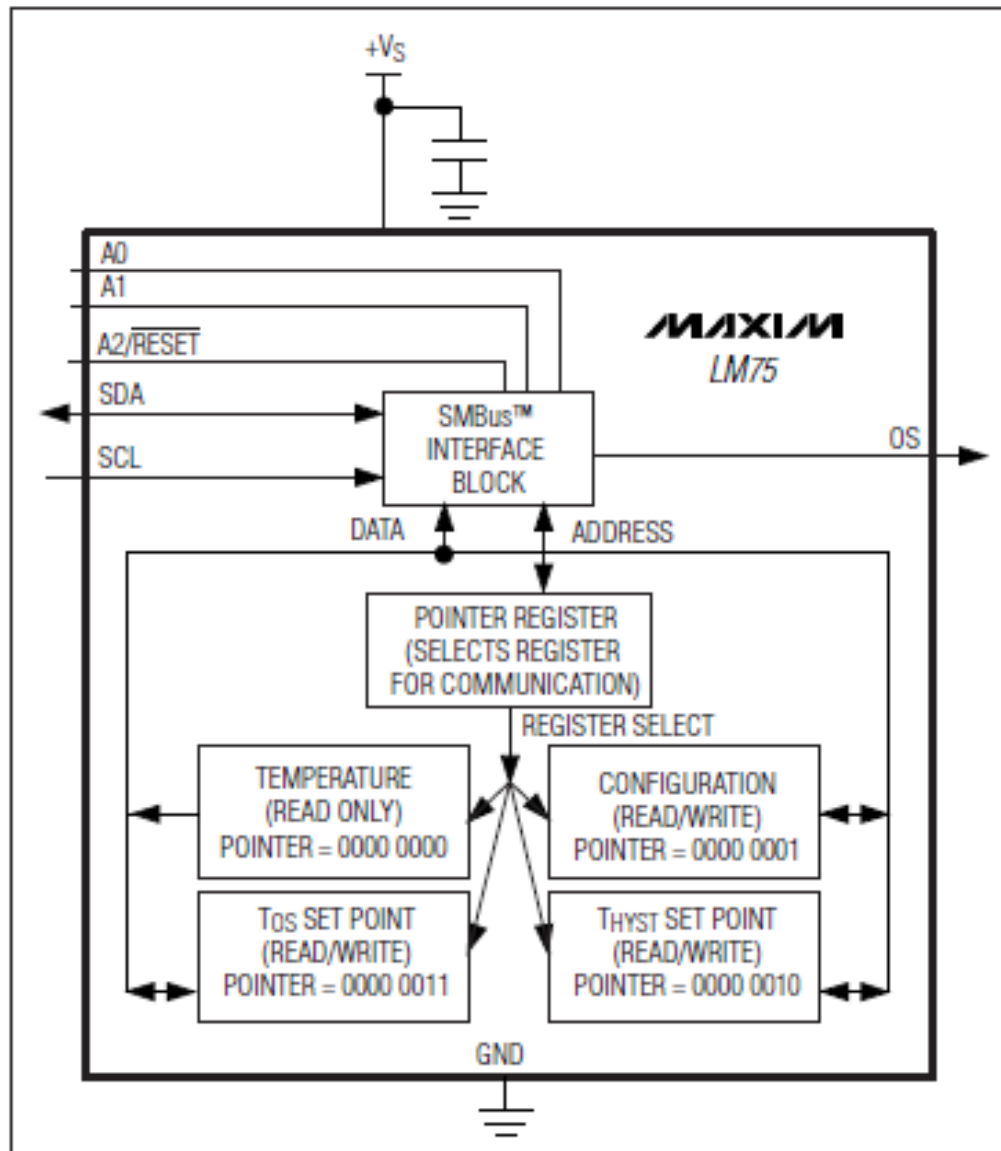
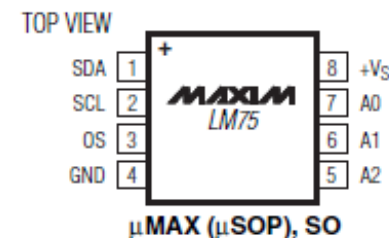


Table 3. Temperature, T_{HYST} , and T_{OS} Register Definition

UPPER BYTE								LOWER BYTE							
D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
Sign bit 1 = Negative 0 = Positive	MSB 64°C	32°C	16°C	8°C	4°C	2°C	1°C	LSB 0.5°C	X	X	X	X	X	X	X

X = Don't care.



PIN	NAME	FUNCTION
1	SDA	Serial-Data Input/Output Line. Open drain. Connect SDA to a pullup resistor.
2	SCL	Serial Clock Input. Open drain. Connect SCL to a pullup resistor.
3	OS	Overtemperature Shutdown Output. Open drain. Connect OS to a pullup resistor.
4	GND	Ground
5	A2	2-Wire Interface Address Input. Connect A2 to GND or +V _S to set the desired I ² C bus address. Do not leave unconnected (see Table 1).
6	A1	2-Wire Interface Address Input. Connect A1 to GND or +V _S to set the desired I ² C bus address. Do not leave unconnected (see Table 1).
7	A0	2-Wire Interface Address Input. Connect A0 to GND or +V _S to set the desired I ² C bus address. Do not leave unconnected (see Table 1).
8	+V _S	Positive Supply Voltage Input. Bypass to GND with a 0.1μF bypass capacitor.

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Accuracy (Six-Sigma)		$-25^{\circ}\text{C} \leq T_A \leq +100^{\circ}\text{C}$	-2.0		+2.0	$^{\circ}\text{C}$
		$-55^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$	-3.0		+3.0	
Accuracy (Three-Sigma) (Note 6)		$-25^{\circ}\text{C} \leq T_A \leq +100^{\circ}\text{C}$	-1.5		+1.5	$^{\circ}\text{C}$
		$-55^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$	-2.0		+2.0	
Resolution				9		Bits
Temperature Conversion Time		(Note 7)		100	300	ms

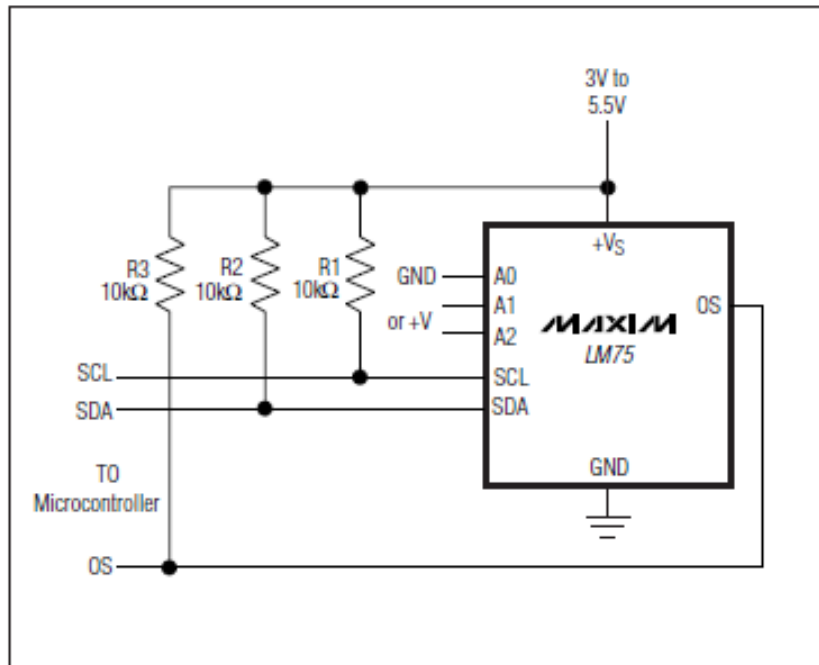


Figure 7. PC Controlled Temperature Sensor

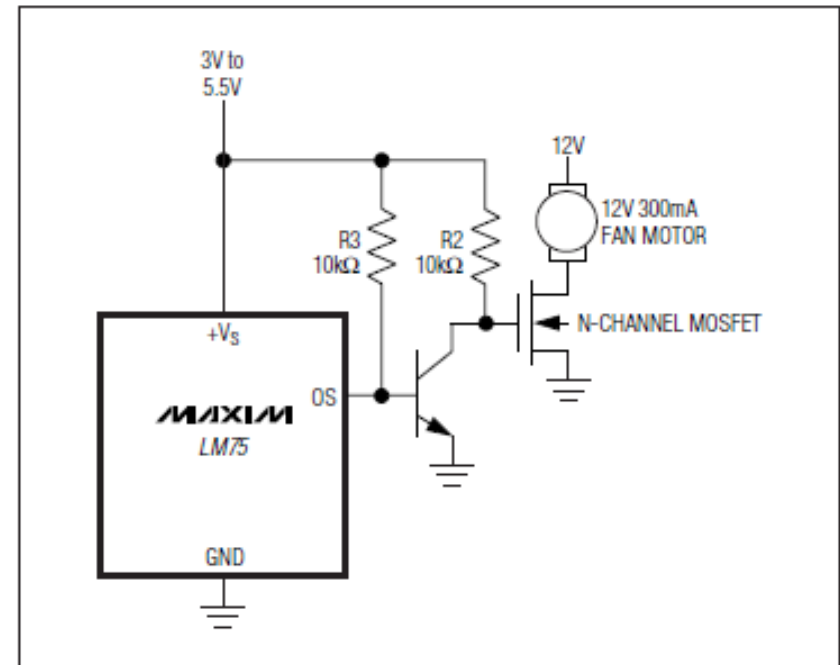


Figure 8. Fan Control